

The Ring R_q : Structure and Fast Arithmetic

Lecture 3 of 6 — Foundations

SEC 595 — Encrypted Computing

- 1 Polynomials Modulo f
- 2 Why $X^n + 1$
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Lecture 1 built $\mathbb{Z}_q$. Lecture 2 built $\mathbb{F}_p[X]$ and the NTT.

This lecture puts the two together and produces the object every scheme in the course computes in:

$$R = \mathbb{Z}[X]/(X^n + 1), \quad R_q = \mathbb{Z}_q[X]/(X^n + 1),$$

with n a power of two.

ML-KEM, ML-DSA, BGV, BFV and CKKS differ enormously in what they do. They all do it here.

- a ciphertext is a pair of elements of R_q ;
- a secret key is an element of R_q with small coefficients;
- noise growth is a statement about $\|\cdot\|_\infty$ on R_q ;
- and the reason any of it is fast is that R_q has a CRT decomposition computable by an NTT.

The lecture has three jobs

- 1 Say precisely what “reduce modulo $X^n + 1$ ” means, and why it is well defined.
- 2 Prove the one inequality $\|ab\|_\infty \leq n\|a\|_\infty\|b\|_\infty$, on which every correctness proof in the FHE half of the course rests.
- 3 Build the negacyclic NTT, and explain exactly why ML-KEM's version of it stops one level short of ML-DSA's.

What each section buys

Section	Where it is used
$\mathbb{Z}[X]/(f)$ and reduction	the definition of every scheme
Negacyclic convolution	every ciphertext multiplication
R_t , R_q , units, zero divisors	encoding; keygen; relinearisation
Expansion factor $\delta_R = n$	every noise bound in BFV, BGV, CKKS
Negacyclic NTT	ML-KEM, ML-DSA, every FHE library
Complete vs. incomplete splitting	FIPS 203 vs. FIPS 204; batching (L4)
Modules R_q^k	MLWE, hence ML-KEM and ML-DSA (L5)

The parameters in play

Small examples use $q = 17$ and $q = 13$, as in Lecture 2.

Real parameters:

- ML-KEM: $q = 3329$, $n = 256$;
- ML-DSA: $q = 8380417$, $n = 256$;
- FHE: $n \in \{2^{12}, 2^{13}, 2^{14}\}$ with $\log q$ in the hundreds.

Lecture 1 built $\mathbb{Z}_n$ from $\mathbb{Z}$: add and multiply as usual, then replace the answer by its remainder on division by n .

$R = \mathbb{Z}[X]/(f)$ is that construction with $\mathbb{Z}[X]$ in place of $\mathbb{Z}$ and a polynomial f in place of the integer n .

Elements: polynomials with integer coefficients. Arithmetic: ordinary polynomial arithmetic, followed by reduction modulo f .

Why f must be monic

For the reduction to be well posed, f must be **monic**.

That is what makes Lecture 2's division algorithm work over $\mathbb{Z}$ and not merely over a field: dividing by the leading coefficient 1 never leaves $\mathbb{Z}$.

$X^n + 1$ is monic. Every f we use is.

No ideal theory is needed here or downstream. What is used is reduction.

Proposition

Let $f \in \mathbb{Z}[X]$ be monic of degree n . Every $a \in \mathbb{Z}[X]$ is congruent modulo f to *exactly one* polynomial of degree $< n$.

Proof. Existence: divide, $a = gf + r$ with $\deg r < n$; then $a - r$ is a multiple of f . *Uniqueness:* if r, r' both have degree $< n$ and $r - r'$ is a multiple of a degree- n polynomial, then $r - r' = 0$. □

So an element is just a coefficient vector

We may, and always will, identify

$$\mathbb{Z}[X]/(f) = \{ a_0 + a_1X + \cdots + a_{n-1}X^{n-1} : a_i \in \mathbb{Z} \}.$$

As a set this is just $\mathbb{Z}^n$: a vector of n coefficients.

What the quotient supplies on top of $\mathbb{Z}^n$ is a **multiplication**: multiply the polynomials, then reduce.

Example — reduction modulo $X^4 + 1$

Take $f = X^4 + 1$, so that $X^4 \equiv -1$.

Reduce

$$X^6 + 3X^5 + 2X^4 + 5X^3 + X + 4$$

to its canonical representative of degree < 4 , and write the answer as a coefficient vector, constant first.

Then reduce the coefficients modulo 17 as well.

Example — the smallest interesting case

Work in $\mathbb{Z}[X]/(X^2 + 1)$, where $X^2 \equiv -1$.

Compute $(2 + 3X)(1 + X)$, reduced to its canonical representative.

Which classical ring have you been computing in, and what plays the role of X ?

The analogy, stated explicitly

	Integers	Polynomials
Ambient ring	$\mathbb{Z}$	$\mathbb{Z}[X]$
Size measure	$ a $	$\deg a$
Division	$a = qn + r, 0 \leq r < n$	$a = gf + r, \deg r < \deg f$
Reduce modulo	n	f , monic
Quotient	$\mathbb{Z}_n$	$\mathbb{Z}[X]/(f)$
Representative	$r \in \{0, \dots, n-1\}$	r with $\deg r < n$
Inverses via	extended Euclid in $\mathbb{Z}$	extended Euclid in $\mathbb{Z}_q[X]$
Field iff	n prime	f irreducible over a field

Every fact we prove about R_q has a companion in Lecture 1 that is already familiar.

$\mathbb{Z}_n$ is a field $\iff n$ prime, $\mathbb{F}_p[X]/(f)$ is a field $\iff f$ irreducible.

We shall choose q and f so that R_q is emphatically **not** a field.

Section 4.3 makes precise why that is a design decision and not an accident.

Forward reference

The two-step construction

$$\mathbb{Z}[X] \longrightarrow \mathbb{Z}[X]/(X^n + 1) \longrightarrow \mathbb{Z}_q[X]/(X^n + 1)$$

gives the same ring in either order: the two reductions act on independent data.

L4 exploits the freedom to reduce coefficients *last*. The BFV product is computed over $\mathbb{Z}$, or over a much larger modulus, and only then pushed back into R_q .

The choice to be justified

We could reduce modulo any monic f .

The whole lattice-cryptography world has settled on

$$f = X^n + 1, \quad n \text{ a power of two.}$$

For such n this polynomial is the $2n$ -th cyclotomic polynomial Φ_{2n} , whose roots over $\mathbb{C}$ are exactly the primitive $2n$ -th roots of unity.

Split the $2n$ -th roots of unity by factoring

$$X^{2n} - 1 = (X^n - 1)(X^n + 1).$$

The roots of $X^n - 1$ are the non-primitive ones; the roots of $X^n + 1$ are the primitive ones.

Consequently

$$R = \mathbb{Z}[X]/(X^n + 1) \cong \mathbb{Z}[\zeta_{2n}],$$

and R is called a **power-of-two cyclotomic ring**.

Four properties make it the choice

In the order they will matter to us:

- **Sparsity:** two nonzero coefficients, so reduction is free — a sign flip and an index shift (Section 3).
- **Norm behaviour:** expansion factor exactly n , the smallest possible for a degree- n ring (Section 5).
- **Fast multiplication:** it splits into linear or low-degree factors over $\mathbb{Z}_q$ whenever q has enough two-adicity (Sections 6 and 7).
- **Automorphisms:** $X \mapsto X^j$ for odd j , forming $\mathbb{Z}_{2n}^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_{n/2}$.

Forward reference

The automorphisms $X \mapsto X^j$ become the ROTATE and CONJUGATE operations of BGV and CKKS.

Combined with the CRT decomposition of Section 7 they give **batching**: n (or $n/2$) independent plaintext slots packed into one ciphertext, operated on in parallel and permuted by the rotations.

That is L4 in its entirety.

The wrap-around rule

In $R = \mathbb{Z}[X]/(X^n + 1)$ the defining relation is

$$X^n \equiv -1, \quad \text{hence} \quad X^{n+r} \equiv -X^r \quad (0 \leq r < n).$$

Multiplication is ordinary polynomial multiplication, followed by folding every degree $\geq n$ back down *with a sign change*.

Proposition

For $a = \sum_{i < n} a_i X^i$ and $b = \sum_{j < n} b_j X^j$ in R , $ab = \sum_{k < n} c_k X^k$ with

$$c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j,$$

both sums over $0 \leq i, j < n$.

Proof. The ordinary product has $d_m = \sum_{i+j=m} a_i b_j$. For $m < n$ the term survives; for $m = k + n$ we have $X^m = X^n X^k \equiv -X^k$. Collect. $\square$

One sign, and the whole NTT has to be rebuilt

modulo $X^n - 1$: $c_k = \sum_{i+j=k} a_i b_j + \sum_{i+j=k+n} a_i b_j$ *cyclic*

modulo $X^n + 1$: $c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$ *negacyclic*

One sign — and it is the reason Section 6 exists.

Example — a negacyclic product by hand

Take $n = 4$, $q = 17$, and

$$a(X) = 1 + 2X + 3X^2 + 4X^3, \quad b(X) = 4 + 3X + 2X^2 + X^3.$$

Compute $a \cdot b$ in $R_{17} = \mathbb{Z}_{17}[X]/(X^4 + 1)$: convolve over $\mathbb{Z}$, fold with the sign, then reduce. What is $\|ab\|_\infty$ in centered representatives?

Example — what a plain NTT would have given

Lecture 2, Exercise 8 asked for $\text{NTT}^{-1}(\text{NTT}(a) \odot \text{NTT}(b))$ with $\omega = 13$ over $\mathbb{F}_{17}$, and observed the answer is *not* $a \cdot b$.

Using the same a and b , compute the *cyclic* product — fold with a $+$ sign — and compare it coefficient by coefficient with the negacyclic one.

Knowing both is knowing the full product

Only the top coefficient agrees — it is the one that receives no wrapped contribution.

From the pair one recovers the unwrapped and wrapped halves of the ordinary product:

$$\text{low half} = \frac{1}{2}(\text{cyclic} + \text{negacyclic}), \quad \text{high half} = \frac{1}{2}(\text{cyclic} - \text{negacyclic}).$$

Which is a special case of CRT, since $X^{2n} - 1 = (X^n - 1)(X^n + 1)$.

Definition

Fix n a power of two, a plaintext modulus $t \geq 2$ and a ciphertext modulus $q \gg t$. Set

$$R = \mathbb{Z}[X]/(X^n + 1), \quad R_t = \mathbb{Z}_t[X]/(X^n + 1), \quad R_q = \mathbb{Z}_q[X]/(X^n + 1).$$

Equivalently $R_t = R/tR$ and $R_q = R/qR$: reduce each of the n coefficients.

Backward reference

Following Lecture 1 we always take the *centered* representative. An element of R_q is a vector of n integers in

$$\left\{ -\left\lfloor \frac{q}{2} \right\rfloor, \dots, \left\lfloor \frac{q-1}{2} \right\rfloor \right\},$$

and $\|a\|_\infty$ is the largest of their absolute values.

We write $[a]_q$ for centered reduction applied coefficientwise.

Two further pieces of notation, fixed now

R_2 denotes the **ternary** distribution on coefficients $\{-1, 0, 1\}$, used for secret keys.

$\mathcal{X}$ denotes the **error distribution**: a discrete Gaussian with $\sigma = 3.2$, truncated at $\beta = 19 \approx 6\sigma$, per the Homomorphic Encryption Standard.

How big these rings are

An element is n independent coefficients, so

$$|R_t| = t^n, \quad |R_q| = q^n.$$

For $q = 17$ and $n = 4$: $17^4 = 83521$.

For ML-KEM: $|R_q| = 3329^{256} \approx 2^{2995}$.

Proposition

Let q be prime. Then $a \in R_q$ is a **unit** if and only if $\gcd(a(X), X^n + 1) = 1$ in $\mathbb{F}_q[X]$; otherwise a is a **zero divisor**.

Proof sketch. If the gcd is 1, extended Euclid in $\mathbb{F}_q[X]$ gives $ua + v(X^n + 1) = 1$, so $u = a^{-1}$. If $g = \gcd$ is nontrivial, write $X^n + 1 = gh$ with $\deg h < n$; then $ah \equiv 0$ while $h \not\equiv 0$. Since R_q is finite, every element is one or the other. $\square$

Example — a zero divisor in R_{17}

Take $n = 4$, $q = 17$. The roots of $X^4 + 1$ over $\mathbb{F}_{17}$ are 2, 8, 9, 15 (Section 7.2).

Split them into two pairs and build the two quadratics $a = (X - 2)(X - 8)$ and $b = (X - 9)(X - 15)$.

Compute ab in R_{17} . Why does this show that neither a nor b can be inverted?

Example — one element that is always a unit

Show that X is invertible in R , and in every R_q , by finding its inverse explicitly.

Which other elements of R have an inverse with all-integer coefficients?

R_q is never a field (once $n \geq 4$)

R_q is a field precisely when q is prime *and* $X^n + 1$ is irreducible over $\mathbb{F}_q$.

The degree of each irreducible factor is the multiplicative order of q in $\mathbb{Z}_{2n}^\times$. For $2n \geq 8$ that group is $\mathbb{Z}_2 \times \mathbb{Z}_{n/2}$, whose exponent is only $n/2$.

So no factor reaches degree n , and there are always at least two of them.

The question is never *whether* R_q decomposes but *how far* — and the standards push it as far as they can.

- **Speed:** a complete or near-complete splitting is what makes the NTT possible; irreducibility would force $O(n^2)$ multiplication.
- **Batching:** the CRT factors of R_t are the SIMD plaintext slots, and an irreducible $X^n + 1$ over $\mathbb{F}_t$ would give exactly one.
- **Composite moduli:** FHE uses $q = \prod_i q_i$, so $\mathbb{Z}_q \cong \prod_i \mathbb{Z}_{q_i}$ (RNS) — filling R_q with zero divisors by construction, and letting the modulus chain be truncated one prime at a time.

The payoff of Lecture 1's warning

Lecture 1 warned that fields are not the goal. Here is why.

Almost every good property of R_q in this course — fast multiplication, SIMD slots, the modulus chain — comes from its **zero divisors**.

The one place their absence is felt is inversion, and schemes are designed to avoid needing it.

Definition

For $a = \sum_{i < n} a_i X^i \in R$, set $\|a\|_\infty = \max_i |a_i|$.

For $a \in R_q$, apply this to the centered representative: $\|a\|_\infty = \|[a]_q\|_\infty$.

The coefficient-wise ℓ_∞ norm on $\mathbb{Z}^n$, transplanted to R by the identification of Section 1.1.

It is not a ring norm

$\|\cdot\|_\infty$ is not multiplicative: $\|ab\|_\infty \neq \|a\|_\infty \|b\|_\infty$ in general.

Quantifying *how badly* it fails to be multiplicative is the single most-used result in the course.

Theorem

For all $a, b \in R = \mathbb{Z}[X]/(X^n + 1)$:

$$(i) \quad \|a + b\|_{\infty} \leq \|a\|_{\infty} + \|b\|_{\infty}, \quad (ii) \quad \|a \cdot b\|_{\infty} \leq n \|a\|_{\infty} \|b\|_{\infty}.$$

Addition is free. Multiplication costs a factor of the ring dimension.

That asymmetry is the whole economics of FHE.

Why they hold — a counting argument

(i) Addition is coefficientwise: $|a_k + b_k| \leq |a_k| + |b_k|$; take the maximum over k .

(ii) By Section 3.1, $c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$. The two sums are disjoint and together range over the pairs with $i + j \equiv k \pmod{n}$. For each of the n values of i there is *exactly one* such j , so c_k is a signed sum of exactly n products:

$$|c_k| \leq \sum_{i=0}^{n-1} |a_i| |b_{j(i)}| \leq n \|a\|_{\infty} \|b\|_{\infty}.$$



Definition

The **expansion factor** of a ring R is

$$\delta_R = \sup_{a,b \neq 0} \frac{\|ab\|_\infty}{\|a\|_\infty \|b\|_\infty}.$$

The theorem says $\delta_R \leq n$ for $R = \mathbb{Z}[X]/(X^n + 1)$.

The bound is attained, so in fact $\delta_R = n$ exactly: make all n signed terms in some c_k agree in sign and have absolute value $\|a\|_\infty \|b\|_\infty$.

Example — the bound is tight

Take $n = 4$ and

$$a = b = 1 + X + X^2 + X^3 \in \mathbb{Z}[X]/(X^4 + 1).$$

Compute ab and $\|ab\|_\infty$, and confirm $\|ab\|_\infty = n \|a\|_\infty \|b\|_\infty$. Which coefficient attains the bound, and why that one?

Then compute the *cyclic* square of the same pair, and say how many of its coefficients hit the bound.

It is a *worst case*. For random a, b the n terms in c_k have random signs, so

$$|c_k| \approx \sqrt{n} \|a\|_\infty \|b\|_\infty, \quad \text{not} \quad n \|a\|_\infty \|b\|_\infty.$$

Libraries use heuristic $\sqrt{n}$ -style bounds for parameter selection and reserve $\delta_R = n$ for provable statements.

Every bound in this course is of the provable kind.

A BFV homomorphic multiplication of ciphertexts with noises v_1, v_2 produces noise of roughly

$$2 t n^2 \|SK\|_\infty \cdot \max(\|v_1\|_\infty, \|v_2\|_\infty).$$

Note the n^2 : the operation runs two ring products in sequence, each paying a factor $\delta_R = n$.

Addition, by contrast, costs only $\|v_1\|_\infty + \|v_2\|_\infty$ — inequality (i), with no factor of n at all.

Why depth, not size, is the budgeted resource

Multiplication *multiplies* the noise; addition only adds it.

So a circuit's **multiplicative depth**, not its gate count, determines how large q must be.

The correctness condition

In BGV at level ℓ , with modulus q_ℓ , decryption is correct provided

$$\|v\|_\infty < \frac{q_\ell}{2t}.$$

Every parameter-selection procedure in the course is the same calculation: bound $\|v\|_\infty$ after the intended circuit using (i) and (ii), then choose q large enough that the inequality still holds at the last level.

That is the entire shape of a correctness proof in this course.

Forward reference

The first place the inequality earns its keep is the fresh-ciphertext noise of BFV, where

$$ct = (\Delta m + PK_0 u + e_1, PK_1 u + e_2)$$

evaluates at the secret to $\Delta m + (-eu + e_1 + e_2 SK)$, and two applications of (ii) bound the bracket.

That derivation — in both its generic $2n\beta^2 + \beta$ and ternary $2n\beta + \beta$ forms, with the numbers for $n = 4096$ — is Section 7 of the BFV lecture. It is not repeated here.

Sections 1–5 complete.

So far: polynomials modulo f , why $X^n + 1$, negacyclic convolution, the rings R_t and R_q , and the expansion factor $\delta_R = n$.

Session 7 resumes at *Roots of unity and the negacyclic NTT*.

The problem

Lecture 2 gave a length- n NTT over $\mathbb{F}_q$ whenever $n \mid q - 1$, and showed that transforming, multiplying pointwise and transforming back computes a product.

Section 3.3 showed *which* product: the **cyclic** one, modulo $X^n - 1$.

That is the wrong ring.

The NTT evaluates at $\omega^0, \dots, \omega^{n-1}$: the n -th roots of unity, which are the roots of $X^n - 1$.

Evaluation at a root of f is precisely the quotient map onto $\mathbb{F}_q[X]/(f)$.

To get $X^n + 1$ we must evaluate at *its* roots instead — the primitive $2n$ -th roots of unity.

The roots of $X^n + 1$ are the odd powers of ψ

Suppose $\mathbb{F}_q$ contains ψ of order exactly $2n$. Then $\omega := \psi^2$ has order n , and the roots are

$$\psi, \psi^3, \psi^5, \dots, \psi^{2n-1} \quad \text{that is} \quad \psi \omega^j, \quad j = 0, \dots, n-1.$$

Each satisfies $(\psi \omega^j)^n = \psi^n \omega^{jn} = \psi^n = -1$, since ψ^n has order 2.

The twist — no new transform is needed

The negacyclic transform we want is

$$\tilde{a}_j = a(\psi \omega^j) = \sum_{i=0}^{n-1} a_i \psi^i \omega^{ij} = \sum_{i=0}^{n-1} (\psi^i a_i) \omega^{ij}.$$

Read the right-hand side: it is the *ordinary* length- n NTT of Lecture 2, applied to the twisted vector $(\psi^i a_i)_i$.

Negacyclic multiplication in five steps

Step	Operation
1. Twist	$a_i \mapsto \psi^i a_i$, and $b_i \mapsto \psi^i b_i$
2. Transform	NTT of length n with $\omega = \psi^2$
3. Pointwise	$\tilde{c}_j = \tilde{a}_j \tilde{b}_j$
4. Inverse	NTT ⁻¹ with ω^{-1} and n^{-1}
5. Untwist	$c_i \mapsto \psi^{-i} c_i$

Proposition

Steps 1–5 compute $a \cdot b$ in $\mathbb{Z}_q[X]/(X^n + 1)$.

Proof sketch. Steps 1–2 compute $a(\psi\omega^j)$ for all j : the image of a under

$$\mathbb{F}_q[X]/(X^n + 1) \longrightarrow \prod_j \mathbb{F}_q[X]/(X - \psi\omega^j).$$

That map is a ring isomorphism (Section 7.1), so pointwise multiplication of images is the image of the product; steps 4–5 invert it. $\square$

What the twist costs

$2n$ scalar multiplications for the twist and n for the untwist, all with precomputed powers of ψ .

In practice the twist is folded into the butterflies of the first NTT layer and costs **nothing at all**.

Appendix A carries out all five steps for $n = 4$, $q = 17$.

Why $2n \mid q - 1$, not $n \mid q - 1$

The construction needs ψ of order $2n$. By Lecture 1, $\mathbb{F}_q^\times$ is cyclic of order $q - 1$ and contains an element of order d iff $d \mid q - 1$. So

$$2n \mid q - 1.$$

An element of order n is not enough: ω alone reaches the roots of $X^n - 1$, and the roots of $X^n + 1$ are simply not in the group it generates.

Since n is a power of two, this says the two-adicity of q is at least $\log_2 n + 1$.

Example — finding ψ and ω for $n = 4$, $q = 17$

Check that $2n \mid q - 1$ here.

Lecture 2 recorded that 3 is a primitive root of $\mathbb{F}_{17}$. Use it to produce a ψ of order 8 and the corresponding ω .

How many valid choices of ψ are there, and what are they?

Theorem

Let q be prime and $X^n + 1 = f_1 f_2 \cdots f_r$ with the f_i pairwise coprime in $\mathbb{F}_q[X]$. Then

$$\mathbb{Z}_q[X]/(X^n + 1) \cong \prod_{i=1}^r \mathbb{F}_q[X]/(f_i),$$

by $a \mapsto (a \bmod f_1, \dots, a \bmod f_r)$, respecting both $+$ and $\times$ componentwise.

Lecture 1's CRT verbatim, with “coprime moduli” becoming “coprime factors”.

All the factors look alike

Every irreducible factor of $X^n + 1$ over $\mathbb{F}_q$ has the same degree n/r , because Frobenius permutes them transitively.

And whenever $q \equiv 1 \pmod{4}$ — which covers every modulus in this course — the factors are binomials:

$$X^n + 1 = \prod_{i=1}^r (X^{n/r} - r_i), \quad r_i \in \mathbb{F}_q.$$

The NTT is the case $r = n$

When $r = n$: every factor is linear, every quotient $\mathbb{F}_q[X]/(X - \psi\omega^j)$ is a copy of $\mathbb{F}_q$, and “pointwise” means scalar multiplication.

When $r < n$: the same transform runs for $\log_2 r$ layers instead of $\log_2 n$, and the pointwise step is multiplication of degree- (n/r) polynomials modulo $X^{n/r} - r_i$.

Proposition

Let q be prime with $q \equiv 1 \pmod{4}$, let $e \geq 2$ be its two-adicity, and let $n = 2^k$. Then $X^n + 1$ factors over $\mathbb{F}_q$ into

$$r = \min(n, 2^{e-1})$$

irreducible factors, each of degree n/r and of the form $X^{n/r} - r_i$.

One formula, and it predicts both standards.

Why the count is what it is

Proof sketch. $\mathbb{F}_q$ contains a primitive 2^j -th root of unity exactly when $j \leq e$. The splitting proceeds by repeatedly applying

$$X^{2m} - c = (X^m - d)(X^m + d) \quad \text{whenever } d = \sqrt{c} \in \mathbb{F}_q.$$

Each layer doubles the number of factors and needs a root of unity of twice the order of the last, so the process stalls after $e - 1$ layers — or earlier, if the factors have already become linear. □

Forward reference

Apply the theorem to R_t rather than R_q . If $X^n + 1$ splits over $\mathbb{F}_t$ into r factors of degree d , then

$$R_t \cong \prod_{i=1}^r \mathbb{F}_{t^d},$$

and a single plaintext carries r independent **slots** that add and multiply in parallel.

For $t \equiv 1 \pmod{2n}$ one gets $r = n$ slots of $\mathbb{F}_t$; the CKKS analogue over $\mathbb{C}$ gives $n/2$ complex slots, because conjugate roots are paired. L4 constructs both, with the slot permutations.

Example — $X^4 + 1$ over $\mathbb{F}_{17}$

Here $q - 1 = 16 = 2^4$, so $e = 4$ and $2n = 8 \mid 16$.

Predict the number and degree of the factors, then find them explicitly.

Check your answer against the sum and the product of the roots — what should each of those equal, and why?

Example — $X^4 + 1$ over $\mathbb{F}_{13}$

Here $q - 1 = 12 = 2^2 \cdot 3$, so $e = 2$ and $8 \nmid 12$.

Predict the number and degree of the factors. Then find them — start from $5^2 \equiv -1$, so that $X^4 + 1 \equiv X^4 - 25$ — and verify by expanding.

Why does neither factor split further?

Same polynomial, same n , different two-adicity

Over $\mathbb{F}_{17}$: four linear factors — complete splitting.

Over $\mathbb{F}_{13}$: two quadratics — incomplete splitting.

The sole difference between the two fields is the two-adicity of the modulus. That miniature is the entire ML-KEM / ML-DSA story.

The same contrast at full size

Exactly this distinction, at $n = 256$, separates the two NIST standards.

ML-DSA: $q = 8380417$, $q - 1 = 2^{13} \cdot 1023$, so $e = 13$ and $r = \min(256, 2^{12}) = 256$ linear factors — a complete 8-layer transform.

ML-KEM: $q = 3329$, $q - 1 = 2^8 \cdot 13$, so $e = 8$ and only $r = \min(256, 2^7) = 128$ quadratic factors — a 7-layer transform, whose pointwise step multiplies linear polynomials modulo $X^2 - \zeta_i$.

Forward reference

The consequences of that one-layer difference — why FIPS 203 specifies a `BaseCaseMultiply` routine where FIPS 204 needs only scalar products, and what $q = 3329$ bought in exchange — are worked out in the ML-KEM lecture, Section 6, and the ML-DSA lecture, Section 5.

When one ring element is not enough

A single ring element is often not enough security, and enlarging n is a clumsy way to get more. The alternative is to take vectors.

Definition

R_q^k is the set of k -tuples $(s_1, \dots, s_k)$ with $s_i \in R_q$, with componentwise addition and componentwise scaling by elements of R_q . It is a free **module** of rank k over R_q .

The same definition as a vector space, with the field replaced by a ring.

A matrix $A \in R_q^{k \times k}$ acts by $(As)_i = \sum_j A_{ij}s_j$, each $A_{ij}s_j$ a negacyclic product.

So As costs k^2 ring multiplications — that is, k^2 pointwise products in the NTT domain, plus the transforms.

$\|s\|_\infty = \max_i \|s_i\|_\infty$, and

$$\|As\|_\infty \leq k n \|A\|_\infty \|s\|_\infty,$$

by applying the expansion bound to each of the k terms of each entry.

The expansion factor of the module is kn — exactly the dimension of the corresponding lattice.

Definition (informal)

Sample $A \leftarrow R_q^{k \times k}$ uniformly, and s, e small from a distribution on R_q^k . An **MLWE sample** is

$$(A, t), \quad t = As + e \in R_q^k.$$

The decision problem is to distinguish $(A, As + e)$ from (A, u) for uniform $u \in R_q^k$.

Two special cases you already know

- $n = 1$: the ring is $\mathbb{Z}_q$ and this is plain **LWE** in dimension k .
- $k = 1$: this is **Ring-LWE** in R_q , and the single equation $t = as + e$ carries n scalar constraints at once.

MLWE interpolates, and the underlying lattice has dimension nk — the collision of notation Lecture 1 warned about, now concrete.

Example — a toy MLWE sample over R_{17}

Take $n = 4$, $k = 2$, $q = 17$, polynomials written as coefficient vectors, constant first, and s, e ternary:

$$A = \begin{pmatrix} (3, 5, 11, 2) & (9, 1, 4, 14) \\ (7, 12, 2, 6) & (15, 8, 10, 1) \end{pmatrix},$$

$$s = \begin{pmatrix} (1, -1, 0, 1) \\ (-1, 0, 1, -1) \end{pmatrix}, \quad e = \begin{pmatrix} (0, 1, -1, 0) \\ (1, 0, 0, -1) \end{pmatrix}.$$

Compute $t = As + e \in R_{17}^2$. How many integers does each of A and t hold, and what is the dimension of the underlying lattice?

ML-KEM's three parameter sets

Parameter set	n	k	lattice dimension nk
ML-KEM-512	256	2	512
ML-KEM-768	256	3	768
ML-KEM-1024	256	4	1024

They differ only in k .

Why n is held fixed — one NTT

Changing n changes the ring, hence the required two-adicity of q , hence q itself, hence the reduction constants, the compression parameters, and the entire transform.

Changing k changes a loop bound.

One arithmetic core — and one certification — covers all three security levels.

Why n is held fixed — granularity

With power-of-two n , the only way to raise security by changing n is to *double* it.

Raising k moves in steps of n , which is the granularity the security levels actually need.

Why n is held fixed — less structure

A rank- k module lattice has fewer subfields and fewer automorphisms to exploit than a degree- nk ring.

MLWE is a hedge against the possibility that Ring-LWE's extra structure turns out to help attackers.

FHE, which *needs* that structure for batching and rotations, takes the opposite bet: $k = 1$ with n in the thousands.

Forward reference

Everything about *hardness* has been deferred.

Why $As + e$ should be indistinguishable from uniform; how MLWE reduces to a worst-case lattice problem; what the approximation factor is and how it depends on q/β — that is L5.

This section supplies only the algebra those statements are written in.

- 1 $\mathbb{Z}[X]/(f)$ for monic f of degree n is polynomials with arithmetic done modulo f : multiply, then reduce. Each class has one representative of degree $< n$, so the ring is $\mathbb{Z}^n$ with a twisted multiplication.
- 2 For n a power of two, $X^n + 1 = \Phi_{2n}(X)$, whose roots are the primitive $2n$ -th roots of unity. It is chosen for sparsity, small expansion factor, NTT-friendliness, and its automorphisms.

- 3 In $\mathbb{Z}[X]/(X^n + 1)$, $X^n = -1$, so multiplication is negacyclic convolution: $c_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$. For the worked pair over $\mathbb{Z}_{17}$ this gives $(1, 0, 16, 13)$, against $(7, 5, 7, 13)$ cyclically.
- 4 R_t and R_q are R with coefficients reduced, taken centered; $|R_q| = q^n$, so $|R_{17}| = 83521$ at $n = 4$. An element is a unit iff it is coprime to $X^n + 1$, and R_q is deliberately not a field: its zero divisors give the NTT, batching, and the RNS modulus chain.

- 5 $\|a + b\|_\infty \leq \|a\|_\infty + \|b\|_\infty$ and $\|ab\|_\infty \leq n\|a\|_\infty\|b\|_\infty$, the latter because each product coefficient is a signed sum of exactly n terms. $\delta_R = n$ is attained, by $a = b = 1 + X + X^2 + X^3$ at $n = 4$. Hence the BFV multiplication growth factor $\approx 2tn^2\|SK\|_\infty$ and the BGV correctness condition $\|v\|_\infty < q_\ell/(2t)$.

- 6 Negacyclic multiplication by NTT: twist by ψ^i with ψ of order $2n$, transform with $\omega = \psi^2$, multiply pointwise, invert, untwist. The requirement is $2n \mid q - 1$.
- 7 For $q \equiv 1 \pmod{4}$ with two-adicity e , CRT decomposes R_q into $r = \min(n, 2^{e-1})$ factors of degree n/r . Over $\mathbb{F}_{17}$, $X^4 + 1$ splits completely; over $\mathbb{F}_{13}$ it has no root. ML-DSA gets 256 linear factors, ML-KEM 128 quadratics.
- 8 R_q^k is a free module of rank k ; an MLWE sample is $(A, As + e)$, and the underlying lattice has dimension nk . ML-KEM varies $k \in \{2, 3, 4\}$ at fixed $n = 256$.

Lecture 4 — Decompositions, Automorphisms, Embeddings

The CRT decomposition of R_t as SIMD slots; the Galois automorphisms $X \mapsto X^j$ as rotations; and the canonical embedding that turns “small coefficients” into geometry.

Everything this lecture deferred to L4.

Example — all five steps, $n = 4$, $q = 17$

Take $\psi = 9$, $\omega = 13$, and the same pair as in Section 3.2:

$$a = 1 + 2X + 3X^2 + 4X^3, \quad b = 4 + 3X + 2X^2 + X^3.$$

Run twist, transform, pointwise, inverse transform, untwist. You will need the powers of ψ and ω , and ψ^{-1} , ω^{-1} , n^{-1} in $\mathbb{F}_{17}$.

Check the result against the hand computation of Section 3.2.

Forward reference

Scale this up and it is ML-KEM.

FIPS 203 never leaves the transformed domain: keys and ciphertexts are *stored* in NTT form, so encryption and decapsulation perform only steps 3 and 5. The same is true of ML-DSA, and of the ciphertext arithmetic in SEAL, OpenFHE and HEAAN.

The coefficient representation of Section 3 is a specification device. The transformed one is what exists in memory.